

Security vulnerabilities in base protocols

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Abstract:

Many of today's networks and applications still rely on protocols developed in the late 20th and early 21st century. The benefit of using these old protocols is that many vendors support them, there's a huge community of engineers who are familiar with the standards, and these protocols stood the test of time.

However, the designers of these protocols focused on functionality first and security second. Fast forward to today: many industries ranging from medical to manufacturing rely on these same protocols that were designed with security as an afterthought. The implications of exploiting the weaknesses of these universal protocols would have enormous ramifications on the world in all sectors.

Our team seeks to find a particular vulnerability in a protocol that is being widely used and find ways of mitigating against these attacks. In particular, we zeroed in on IPv6 protocols as they are up and coming in terms of widespread use in addition to being implemented very similarly to IPv4 protocols. The reasoning for this is to ease the transition, but as a result IPv6 protocols carry some of the same vulnerabilities as IPv4, and without as many tried and true mitigation methods.

Responsibilities

Our goal in this project is to select a widely used internet protocol and implement mitigations to vulnerabilities that exist in the inherent design of the protocol. Therefore, we're looking for those protocols with confidentiality, integrity, and/or availability issues. We plan on looking at existing solutions and either iterating on their design and/or implement our own.

Deliverables Timeline

February 7: Literature Review Due

February 16: Proposal Due

February 28: Protocols evaluated and one selected

March 6: Vulnerable protocol identified

March 13: Solutions researched and identified

March 27: Midterm Report Due

May 1: Final Deliverables Due

Results

Overview

We chose to analyze the neighbor discovery protocol (NDP), which is part of the IPv6 suite. This protocol serves to gather information from a network to configure communication with other hosts across networks. By using several different message types, a client can build a mapping of nearby machines and determine how to best send information to an intended recipient. One part of NDP we would like to look at is the link-layer mapping. This is very similar to the Address Resolution protocol of IPv4 and thus is also vulnerable to the same attacks, most famously ARP spoofing.

With NDP comes 5 different ICMPv6 message types:

- Router Solicitation
- Router Advertisement
- Neighbor Solicitation
- Neighbor Advertisement
- Redirect

One particular attack we are looking at is NDP spoofing. Using router advertisement and neighbor advertisement messages, anyone on the link can advertise their own MAC address to IPv6 address mapping and poison the cache of the target host.

Design

Our goal is to build a tool that helps to detect an attack, generate alerts, and compose log messages that can be easily digested by a logging solution. We do not intend to block these false advertisements. The goal is to build a solution that just detects an attack.

A solution for this issue exists in the form of NDPmon. This tool is quite large and does more than just alert on address resolution errors. We are developing a simpler tool as a base for future alerting on IPv6 messages and possibly IPv4 messages as well. In addition, we are implementing our simpler tool as a system level process so that it can run invisibly in the background for long periods of time. Our solution covers both Windows and Linux.

Our program uses the Scapy library to dissect the packets to examine the headers. We can either use the built in Linux command for getting the mapping of IPv6 addresses to MAC addresses or we can wait until someone advertises to store it. The program will wait for NDP advertisements and will try to check if it has seen it in the past. If it is new, then it assumes that the message is okay. It is important to note here that there is no authentication of the advertisement message.

Additionally, we have done some preliminary research into Secure NDP and MACSec. These 2 algorithms are designed to use public key cryptography to solve authentication and possibly privacy issues with link layer protocols. Secure NDP is meant only for NDP while MACSec is supposed to be more generic.

Secure NDP does not have nearly the widespread adoption that MACSec has. There are some implementations online, but for the most part, not many systems have this feature by default. On the other hand, MACSec does have an implementation in the Linux kernel and is adopted by the major networking hardware vendors.

Since Secure NDP is not as widespread we decided to implement parts of SeND. Specifically, we chose to implement cryptographically generated addresses. We built this using python and followed the IETF RFC 3972. This program generates the addresses by taking in a DER encoded certificate (that includes the public key), 3-bit security value, and a 64-bit IPv6 prefix.

As an overview, this program takes in a certificate then creates a hash of the certificate combined with the 64-bit prefix. The security value is used to add additional security by forcing the program to continually hash. The loop stops when $16 * \text{security value}$ number of zeroes have appeared in the leftmost bits of our hash.

Literature Review

IPv6 Security Threats and Possible Solutions [1]

This paper surveys various vulnerabilities that exist in IPv6 and looks at shared vulnerabilities with IPv4. IPv6 shares some issues with IPv4 such as sniffing attacks and denial of service attacks. However, the more interesting attacks are the ones more specific to IPv6. A good example is a reconnaissance attack, which allows the attacker to use an IPv6 multicast address to identify resources on the network rather than having to scan the entire subnet since IPv6 subnets are extremely large. The paper lists possible solutions such as using a firewall and an intrusion detection system.

Review of Security Vulnerabilities in the IPv6 Neighbor Discovery Protocol [2]

This paper discusses some security vulnerabilities in the Neighbor Discovery Protocol (NDP) of IPv6. NDP is the equivalent of ARP for IPv6, and the paper highlights three possible attacks using NDP: MiTM, DoS, and ICMPv6 flooding. The MiTM attack can be performed by a malicious node joining the network and responding to a multicast NS message with an NA message with the solicited and override flags set. The DoS attack can be performed by a malicious node preventing a new node from entering the network by preventing it from picking a valid IPv6 address. Flooding attacks can be performed by sending spoofed RA messages, which will result in traffic being improperly routed.

Security vulnerabilities, attacks and countermeasures in wireless sensor networks at various layers of OSI reference model: A Survey [3]

This paper is focused primarily on delineating what sorts of vulnerabilities wireless sensor networks (WSN) are most vulnerable to. The paper steps through each layer of the OSI model to find vulnerabilities at each layer. In regards to IP/TCP layer vulnerabilities, the paper identified IP hijacking, IP spoofing, SYN floods, and smurf attacks as the easiest vectors for attack. To best defend against these attacks, the paper suggested geographic aware-protocols to include location in the protocol, and implementing firewalls to lessen malicious packets. However the main point of the paper was to find attack vectors, rather than defenses against them.

Multicast security: a taxonomy and some efficient constructions [4]

This paper is concerned mainly with finding a midpoint solution to the problem of adding security to multicast communication. It does so by improving on previous literature for group key revocation and adding source authentication to message authentication codes (MACs). This is done in an effort to cut down on the prohibitive overhead associated with adding authentication to protocols.

Smart Grid Security: Threats, Vulnerabilities and Solutions [5]

This paper goes over the threats, vulnerabilities, and solutions that come with intelligent grid infrastructure. Securing protocols becomes a top priority as they are susceptible to DOS attacks that can bring down infrastructure. They tackle this problem by ensuring robust authentication protocols are installed when communicating in between smart grids.

Biographies of Investigators

Ari Bender-Long

Ari Bender-Long holds a Bachelor's degree in Computer Science and is currently pursuing a Master's degree in Computer Science with a specialization in Information Assurance. He is a recipient of the NSF's CyberCorps Scholarship for Service and has completed an internship at Copart as a member of the application security team. At Copart, Ari performed both manual and automated web application security testing using tools such as Burp Suite and AppSpider and created Python scripts for automated internal codebase policy enforcement. His related coursework includes Language-Based Security, Network Security, Information Security. He has experience with Python and Java, and enjoys popping boxes on Hack The Box and participating in CTFs in his spare time.

Xavier Cadot

Xavier Cadot is in his second year of pursuing his Master's in Computer Science with a specialization in Information Assurance. He is a recipient of the CyberCorps Scholarship for Service and will graduate Fall 2020. This past summer he was at the University of Washington Applied Physics Lab developing software for a sonar side array that is designed to find unexploded ordinances on the ocean floor. The first time Xavier was part of INSuRE, he participated in the Counter Drone project in which he developed Bash scripts to deauthenticate drones from their controllers using a Raspberry Pi and Aircrack-ng. He has recently acquired the Security Plus certification.

Andrew Le

Andrew Le holds a Bachelor's degree in computer science from UT Dallas and is currently pursuing a Master's degree in Computer Science with a specialization in Information Assurance. He is a part of the CyberCorps: Scholarship for Service program and will be graduating in December 2020. He has experience developing software in Python across various teams in the Department of Defense as an intern over multiple summers. His interests and side projects include infrastructure management, system administration, and network engineering - general security engineering/devops based topics.

Alejandro Mestanza

Alejandro Mestanza holds a Bachelor's degree with honors in Computer Science and a minor in Mathematics and is currently in the last year of pursuing his Master's degree in Computer Science with a focus in Information Assurance. He is also a recipient of the CyberCorps Scholarship for Service. His previous work experience includes a software development internship at a financial software development company. He was also a global security intern at Lawrence Livermore National Labs where he developed an interest in defending SCADA networks. He has also done work for Texas Instruments involving hardware testing through a joint operation with the University of Texas at Dallas. His personal projects range from hardware tinkering to competing in Hackathons and CTFs.

Bibliography

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[2]

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[3] <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8305855>

[4] <https://ieeexplore.ieee.org/abstract/document/751457/metrics#metrics>

[5] http://www.aloul.net/Papers/faloul_icsgce12.pdf